

BLUESTOCK FINTECH

MUTUAL FUND ANALYTICS

DASHBOARD

USING PYTHON, SQLITE, AND POWER

BI

Bluestock Mutual Fund Capstone Project

Submitted in Partial Fulfillment of the Requirements
for the Successful Completion of the Capstone Program

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Technologies Used

Python | SQLite | Power BI | GitHub

Project Duration

June 2026

Academic Year

2025-2026

CERTIFICATE

This is to certify that Ms. Nandini has successfully completed the project titled “Mutual Fund Analytics Dashboard using Python, SQLite, and Power BI” as part of the Bluestock Mutual Fund Capstone Project.

The project involved end-to-end implementation of the data analytics lifecycle, including data collection, cleaning, exploratory data analysis, performance analytics, advanced risk analysis, and the development of an interactive Power BI dashboard.

The work presented in this report is an original effort carried out by the student under the guidance and support of the Bluestock Team. The project has been completed satisfactorily and fulfills the requirements of the capstone program.

Date: _____

Place: _____

Project Mentor
Bluestock Team

Candidate Signature
Nandini

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Bluestock Fintech for providing me with the opportunity to work on this capstone project, “Mutual Fund Analytics Dashboard using Python, SQLite, and Power BI.”

This project has been an invaluable learning experience that allowed me to apply concepts of data analytics, database management, visualization, and business intelligence to a real-world financial use case.

I extend my heartfelt thanks to the Bluestock Team and mentors for their continuous guidance, support, and valuable feedback throughout the project. Their encouragement and insights helped me overcome challenges and successfully complete the project.

I would also like to acknowledge the developers and communities behind the open-source tools and technologies used in this project, including Python, Pandas, SQLite, Jupyter Notebook, Power BI, and GitHub.

Finally, I am grateful to my family and friends for their constant motivation and support during the completion of this project.

Nandini
June 2026

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EXECUTIVE SUMMARY

The Mutual Fund Analytics Dashboard Project was undertaken to analyze mutual fund industry data and transform it into meaningful insights using data analytics and business intelligence techniques. The project aimed to understand fund performance, investor behavior, and market trends through a structured analytical approach.

The project involved the complete data analytics lifecycle, beginning with data ingestion and cleaning, followed by database creation using SQLite. Exploratory Data Analysis (EDA) was performed to identify patterns and trends in NAV movements, SIP inflows, AUM growth, investor demographics, and portfolio allocations. Advanced performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Value at Risk (VaR), and Maximum Drawdown were calculated to evaluate mutual fund performance and risk characteristics.

An interactive four-page Power BI dashboard was developed to provide stakeholders with an intuitive platform for exploring industry trends, fund performance, investor analytics, and SIP market insights. Additional advanced analytics, including investor cohort analysis, SIP continuity assessment, sector concentration measurement, and a rule-based fund recommendation system, further enhanced the project's analytical capabilities.

The successful completion of this project demonstrates practical proficiency in Python, SQL, Power BI, and data visualization while highlighting the role of data-driven decision-making in the financial services domain. The insights generated through this project can assist investors and analysts in making more informed investment decisions.

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

The mutual fund industry generates a vast amount of data related to fund performance, investor transactions, assets under management, and market trends. However, raw financial data alone does not provide meaningful insights for investors and analysts. There is a need for an integrated analytical solution that can process, analyze, and visualize this data to support informed investment decisions.

This project addresses this challenge by developing an end-to-end Mutual Fund Analytics Dashboard using Python, SQLite, and Power BI. The solution transforms raw datasets into actionable insights through data cleaning, performance evaluation, risk analysis, and interactive business intelligence reporting.

Project Objectives

The key objectives of this project are:

- To collect and integrate mutual fund datasets from multiple sources.
- To perform data cleaning, validation, and transformation using Python.
- To design and implement a SQLite database for efficient data storage and retrieval.
- To conduct Exploratory Data Analysis (EDA) to identify trends and patterns in mutual fund data.
- To evaluate fund performance using metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- To perform advanced analytics including VaR, CVaR, investor cohort analysis, SIP continuity analysis, and sector concentration analysis.

- To develop an interactive Power BI dashboard for visualizing industry trends, fund performance, and investor behavior.
- To provide meaningful insights and recommendations that support data-driven investment decisions.

DATA SOURCES

The project utilized multiple datasets related to the mutual fund industry to perform comprehensive analysis and build an interactive dashboard. These datasets provided information on fund characteristics, investor behavior, market trends, and portfolio composition.

Datasets Used

1. Fund Master

Contains the basic details of mutual fund schemes, including AMFI code, fund house, scheme name, category, benchmark, expense ratio, risk category, and fund manager information.

2. NAV History

Provides historical Net Asset Value (NAV) data for mutual fund schemes. This dataset was used to calculate daily returns, CAGR, rolling performance metrics, and risk measures.

3. AUM by Fund House

Contains Assets Under Management (AUM) information for different fund houses over time. It was used to analyze industry growth and compare fund houses.

4. Monthly SIP Inflows

Includes monthly Systematic Investment Plan (SIP) inflow data, active SIP accounts, and SIP-related growth indicators. This dataset helped identify investment trends.

5. Category-wise Inflows

Provides net inflows and outflows across various mutual fund categories, enabling the analysis of investor preferences.

6. Industry Folio Count

Contains data related to the number of folios across the mutual fund industry and was used to study investor participation trends.

7. Scheme Performance

Includes return metrics, risk measures, AUM, expense ratios, ratings, and other indicators used for performance evaluation and fund ranking.

8. Investor Transactions

Provides transaction-level information such as transaction type, investment amount, state, city, age group, gender, and KYC status. This dataset was used for investor analytics.

9. Portfolio Holdings

Contains portfolio composition details, including stock holdings, sector allocation, and weight percentages, which supported diversification and concentration analysis.

10. Benchmark Indices

Includes benchmark index values used for comparative analysis and evaluation of mutual fund performance against market indices.

Data Source Summary

A total of 10 datasets were integrated and analyzed throughout the project. These datasets formed the foundation for ETL processes, exploratory analysis, performance analytics, advanced risk analysis, and the development of the Power BI dashboard.

ETL DESIGN (Extract, Transform, Load)

The ETL (Extract, Transform, Load) process formed the backbone of this project by ensuring that raw mutual fund datasets were converted into clean, structured, and analysis-ready data.

1. Extract

The extraction phase involved collecting data from multiple CSV datasets related to the mutual fund industry. A total of ten datasets were imported into Python using the Pandas library. In addition, live NAV data was fetched using the Mutual Fund API to supplement the historical records.

Extracted Data Sources

- Fund Master
- NAV History
- Scheme Performance
- Investor Transactions
- AUM by Fund House
- Monthly SIP Inflows
- Category-wise Inflows
- Industry Folio Count
- Portfolio Holdings
- Benchmark Indices

2. Transform

During the transformation phase, several data cleaning and validation techniques were applied to improve data quality and consistency.

Transformation Activities

- Converted date columns into standard datetime format.
- Removed duplicate records.
- Handled missing values using appropriate methods.
- Standardized transaction types and categorical values.
- Validated numerical fields such as NAV, returns, and transaction amounts.
- Checked expense ratio ranges and risk categories.
- Created derived metrics including daily returns and rolling statistics.

These transformations ensured that the datasets were reliable and suitable for further analysis.

3. Load

After cleaning and transformation, the processed datasets were stored in a SQLite database designed using a star schema structure. The data was loaded using SQLAlchemy and Pandas, enabling efficient querying and retrieval for analytical purposes.

Database Components

- **Dimension Tables**
 - Fund Dimension
 - Date Dimension
- **Fact Tables**
 - NAV Fact
 - Transactions Fact
 - Performance Fact
 - AUM Fact

The cleaned datasets were also saved in the `data/processed` folder for reuse in subsequent stages of the project.

ETL Outcome

The ETL process successfully transformed raw mutual fund data into structured and analysis-ready datasets. This foundation enabled exploratory analysis, performance evaluation, advanced analytics, and the development of an interactive Power BI dashboard.

PERFORMANCE ANALYSIS

Performance analysis was conducted to evaluate the return-generating ability and risk-adjusted performance of mutual fund schemes. Various financial metrics were calculated using historical NAV data and scheme performance indicators to compare the effectiveness of different funds.

Performance Metrics Used

1. Compound Annual Growth Rate (CAGR)

CAGR was used to measure the annualized growth rate of mutual fund schemes over the investment period. It provided a standardized method to compare long-term fund performance.

2. Sharpe Ratio

The Sharpe Ratio was calculated to assess risk-adjusted returns by measuring excess returns generated for each unit of total risk undertaken by the fund.

3. Sortino Ratio

Unlike the Sharpe Ratio, the Sortino Ratio considered only downside volatility, providing a better understanding of negative return risk.

4. Alpha

Alpha measured the excess return generated by a fund relative to its benchmark after accounting for market movements. Positive alpha indicated superior fund management performance.

5. Beta

Beta evaluated the sensitivity of a mutual fund's returns to changes in benchmark returns. Higher beta values reflected greater market-related risk.

6. Maximum Drawdown

Maximum Drawdown measured the largest decline from a fund's peak value to its lowest point, indicating potential downside risk during adverse market conditions.

7. Fund Scorecard

A composite scorecard was developed by combining multiple performance indicators, including returns, Sharpe Ratio, Alpha, expense ratio, and drawdown measures. This helped rank mutual funds based on overall performance and risk efficiency.

Key Findings

- Funds with higher Sharpe Ratios delivered better risk-adjusted returns.
- Positive alpha values indicated that several schemes outperformed their benchmarks.
- Lower maximum drawdowns reflected stronger resilience during market downturns.
- The composite scorecard enabled the identification of top-performing funds by balancing both return and risk considerations.

Performance Analysis Outcome

The analysis provided a comprehensive framework for evaluating mutual funds beyond absolute returns. By incorporating multiple risk and return measures, investors can make more informed decisions aligned with their investment objectives and risk appetite.

ADVANCED ANALYTICS

To enhance the analytical capabilities of the project, advanced techniques were applied to evaluate risk, investor behavior, portfolio concentration, and investment recommendations. These analyses provided deeper insights beyond traditional performance measures.

1. Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Historical Value at Risk (VaR) at a 95% confidence level was calculated using the daily return distribution of each mutual fund scheme. VaR estimates the maximum expected loss under normal market conditions.

Conditional Value at Risk (CVaR) was also computed to measure the average loss beyond the VaR threshold, thereby providing a better understanding of extreme downside risk.

2. Rolling Sharpe Ratio Analysis

A 90-day Rolling Sharpe Ratio was calculated to study how the risk-adjusted performance of selected funds changed over time. This analysis helped identify periods of strong performance as well as periods of increased volatility.

3. Investor Cohort Analysis

Investors were grouped based on the year of their first transaction. The analysis evaluated average investment amounts, cumulative investments, and investment preferences across different cohorts, revealing behavioral differences among investor groups.

4. SIP Continuity Analysis

Systematic Investment Plan (SIP) transactions were examined to assess investment discipline. Investors with average gaps exceeding 35 days between SIP transactions were

identified as potentially at risk of discontinuing their investments.

5. Fund Recommendation System

A simple rule-based recommendation system was developed to suggest suitable mutual funds based on an investor's risk appetite. The recommender utilized risk grades and Sharpe Ratios to identify the top-performing schemes for Low, Moderate, and High-risk investors.

6. Sector Concentration Analysis (HHI)

The Herfindahl–Hirschman Index (HHI) was used to measure portfolio concentration. Higher HHI values indicated greater concentration within specific sectors, whereas lower values suggested better diversification.

Key Insights

- Certain schemes exhibited higher VaR and CVaR values, indicating greater downside risk.
- Rolling Sharpe analysis highlighted fluctuations in risk-adjusted performance over time.
- Early investor cohorts contributed higher cumulative investments.
- SIP continuity analysis identified investors who may require retention strategies.
- Sector concentration varied significantly across funds, emphasizing the importance of diversification.

Advanced Analytics Outcome

The incorporation of advanced analytical techniques enabled a more comprehensive assessment of mutual fund performance, investor behavior, and portfolio risk. These insights strengthened the decision-making capabilities of the overall analytics framework.

DASHBOARD SCREENSHOTS

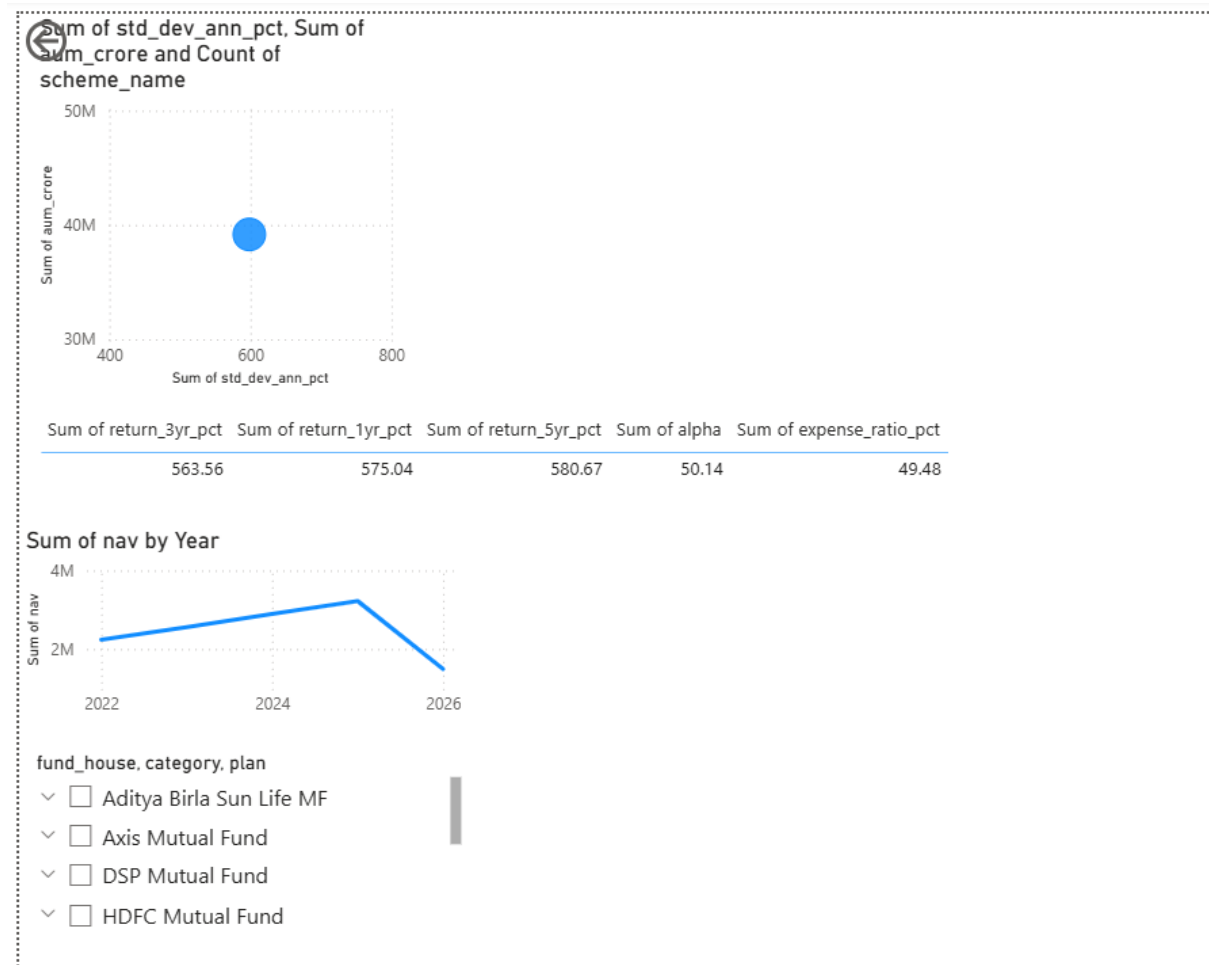


Figure 1: Overview Dashboard

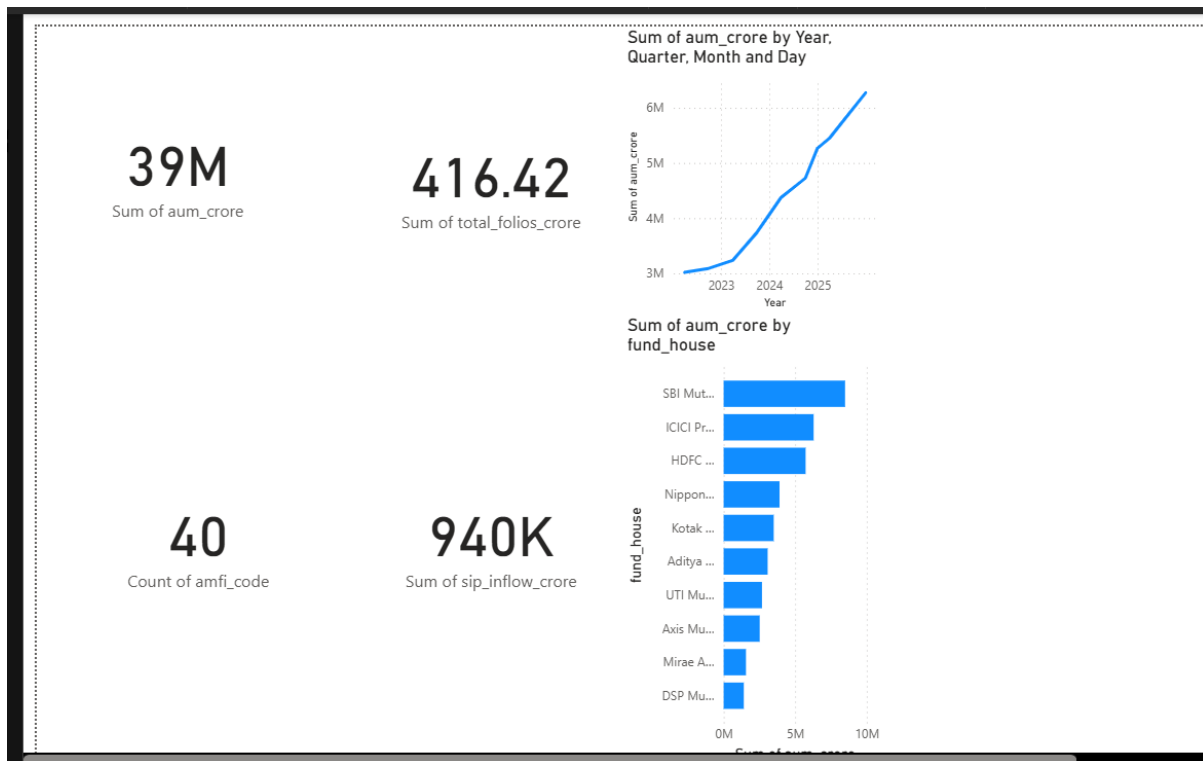


Figure 2: Performance Analytics Dashboard

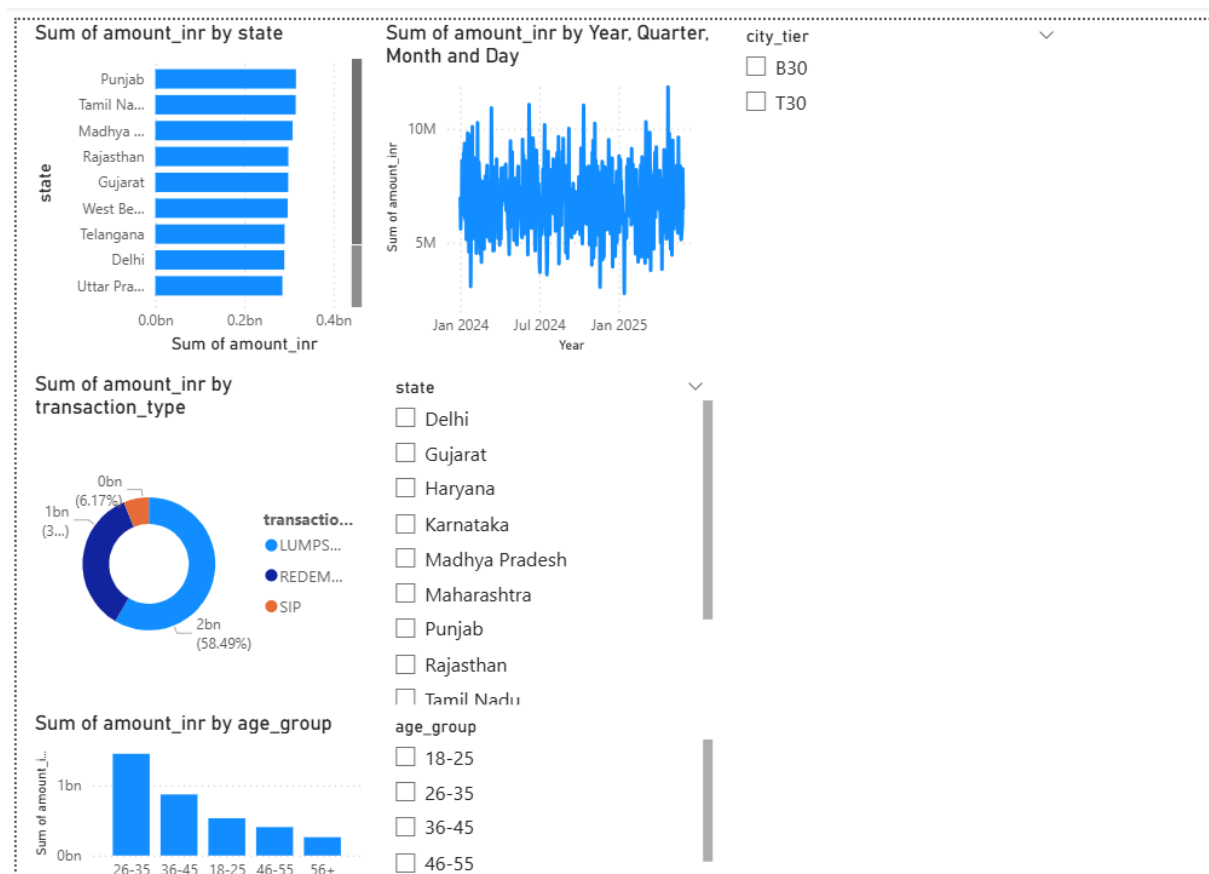


Figure 3: Performance Analytics Dashboard

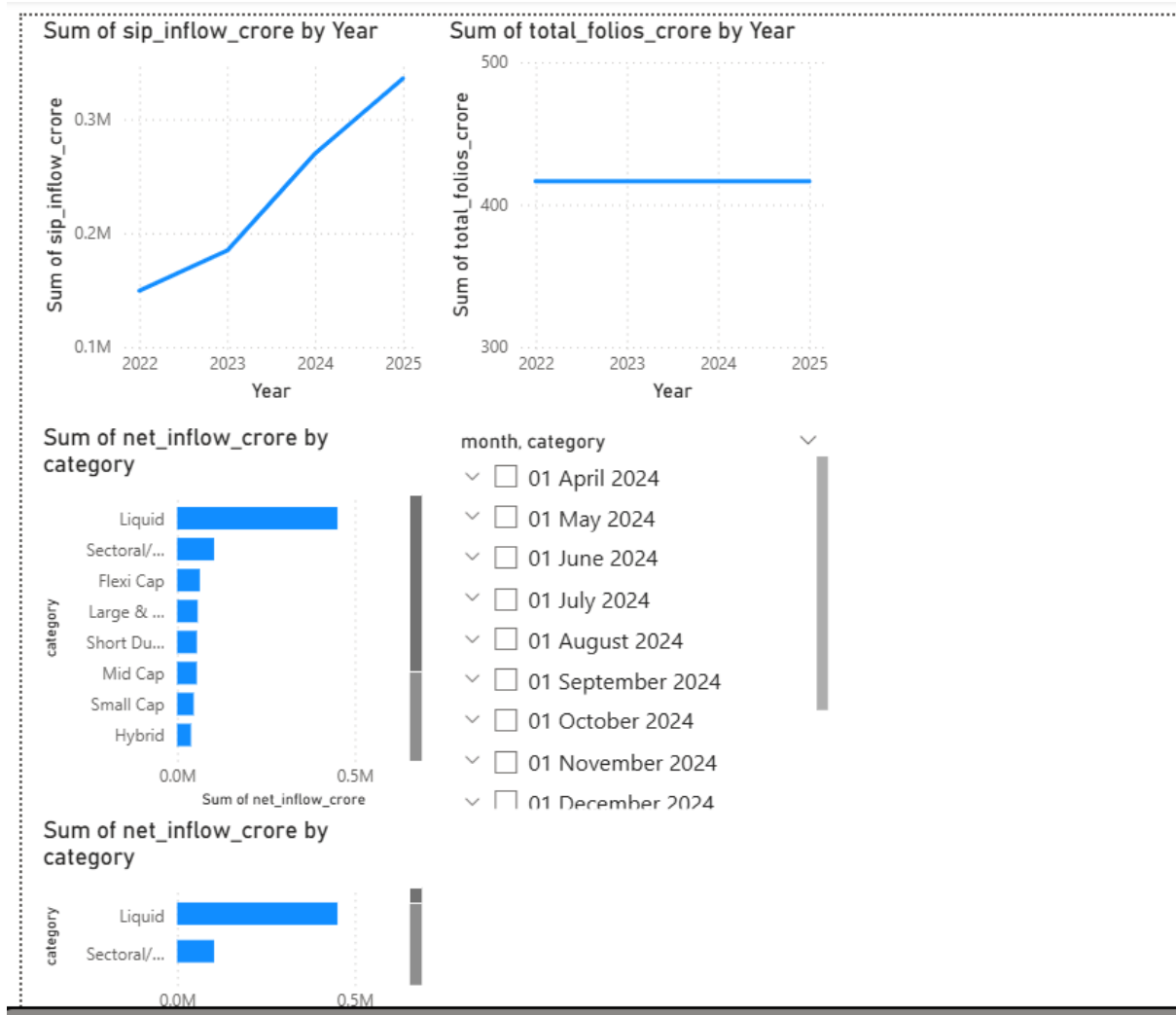


Figure 4: Performance Analytics Dashboard

KEY FINDINGS

The analysis conducted throughout this project provided several valuable insights into the mutual fund industry and investor behavior.

Major Findings

- Mutual fund NAVs exhibited a long-term upward trend, indicating consistent growth potential despite short-term market fluctuations.
- SIP inflows increased significantly from 2022 to 2025, reflecting growing investor awareness and participation.
- Equity-oriented mutual funds attracted the highest net inflows compared to other categories.
- Investors aged between 26 and 35 years constituted the largest share of investment activity.
- T30 cities contributed a higher proportion of investments than B30 cities.
- Funds with higher Sharpe Ratios demonstrated superior risk-adjusted performance.
- Advanced risk measures such as VaR and CVaR highlighted differences in downside risk across schemes.
- Sector concentration varied considerably among funds, emphasizing the importance of diversification.

LIMITATIONS

Although the project achieved its objectives successfully, certain limitations should be acknowledged:

- The analysis was based on historical datasets and may not fully capture future market conditions.
- Real-time market integration was limited to selected API-based NAV retrieval.
- The recommendation system was rule-based and did not incorporate machine learning techniques.
- Benchmark assumptions and proxy risk-free rates may differ from actual market conditions.
- Some datasets represented a limited sample of schemes and investor transactions.

RECOMMENDATIONS

Based on the findings of this project, the following recommendations are proposed:

- Integrate real-time APIs to enable automated updates and live monitoring.
- Develop a machine learning-based recommendation engine for personalized fund suggestions.
- Incorporate predictive analytics for forecasting fund performance.
- Expand the dashboard with additional benchmarking and scenario analysis capabilities.
- Implement automated report generation and dashboard refresh mechanisms.

CONCLUSION

The Mutual Fund Analytics Dashboard project successfully demonstrated the complete data analytics lifecycle, from data ingestion and cleaning to advanced analytics and business intelligence reporting. Using Python, SQLite, Jupyter Notebook, and Power BI, raw mutual fund datasets were transformed into meaningful insights that support informed investment decision-making.

The project not only enhanced the understanding of mutual fund performance and investor behavior but also showcased practical skills in data engineering, analytics, visualization, and dashboard development. The resulting solution provides a comprehensive and user-friendly platform for exploring mutual fund trends and evaluating investment opportunities.

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